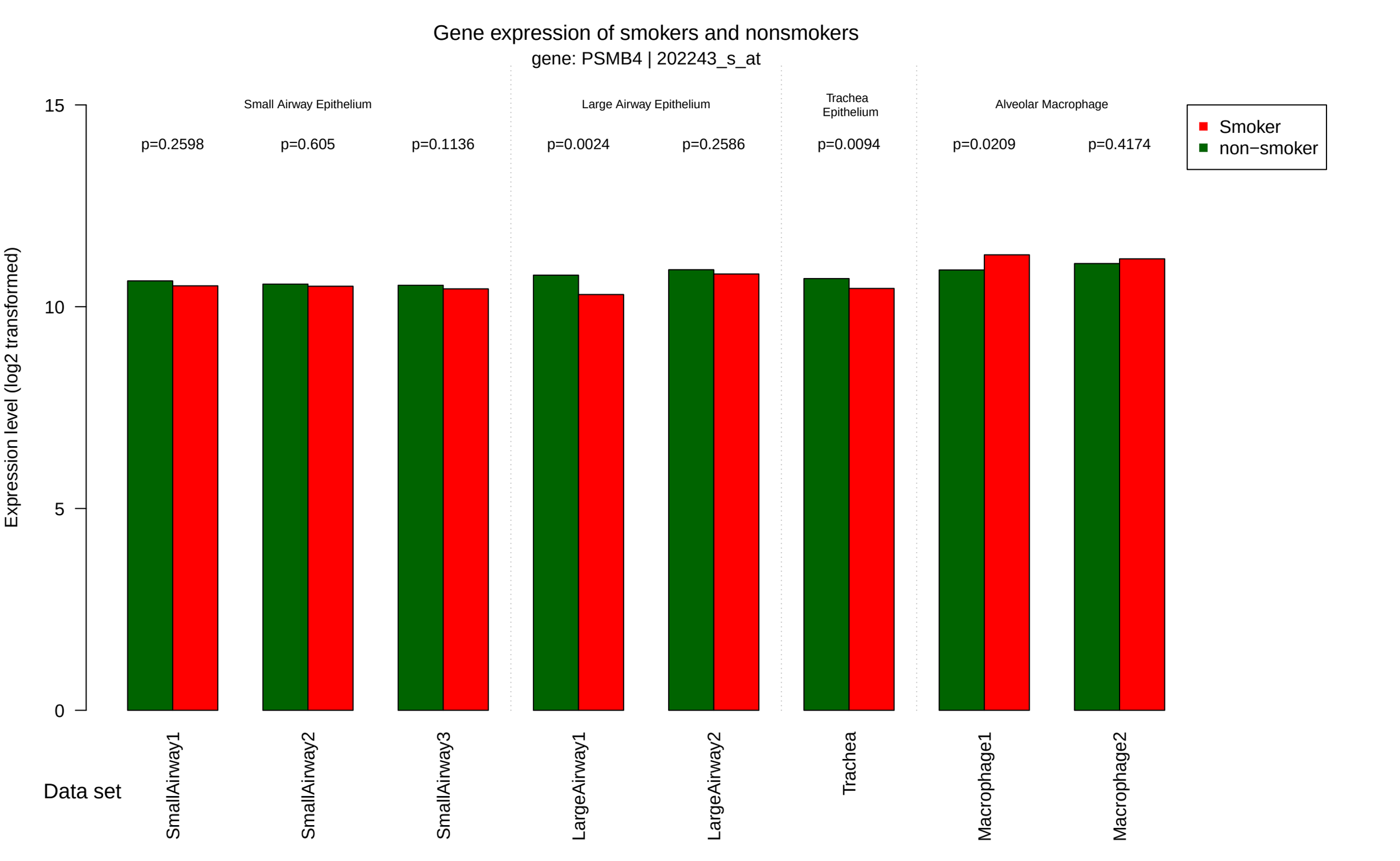
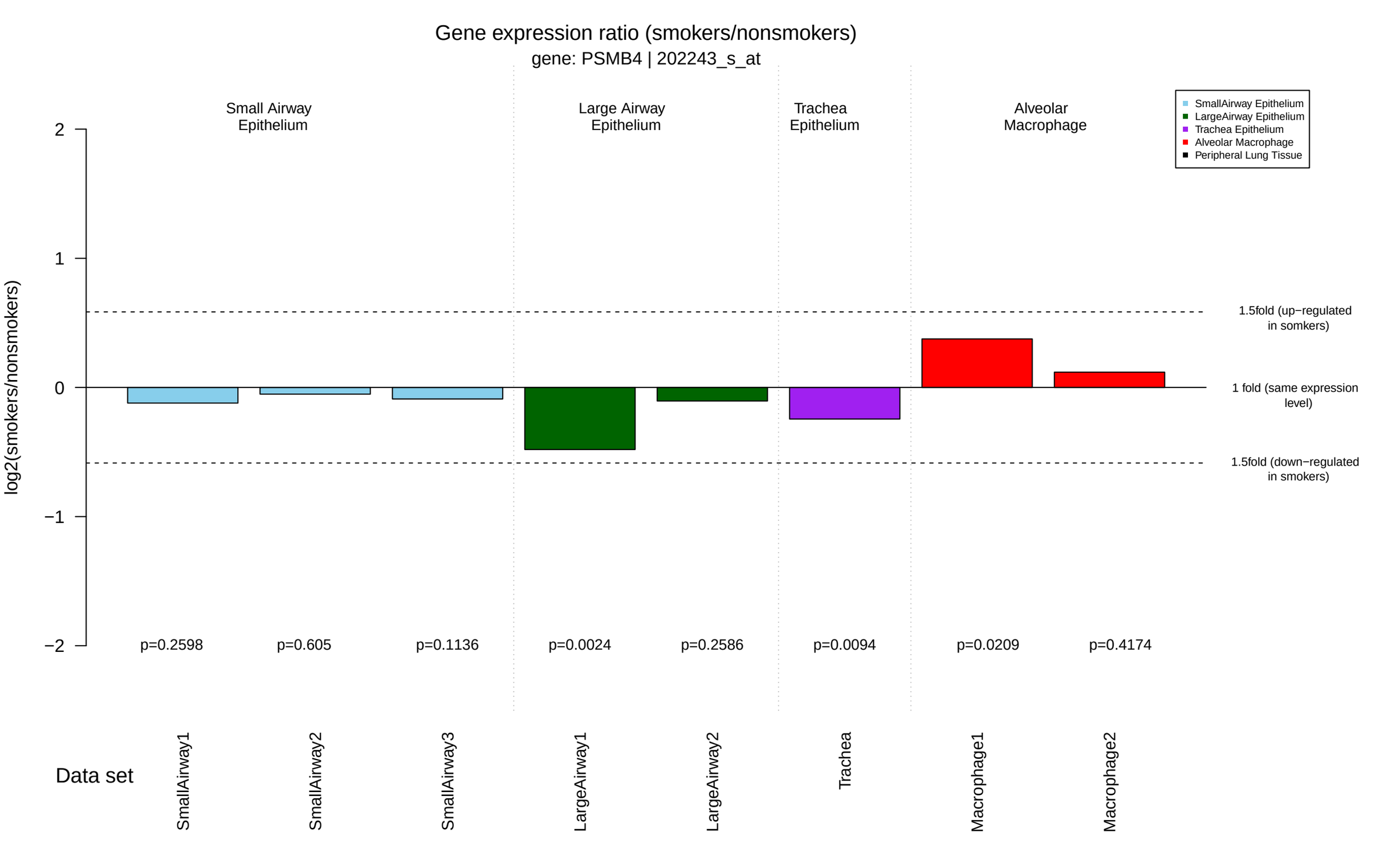
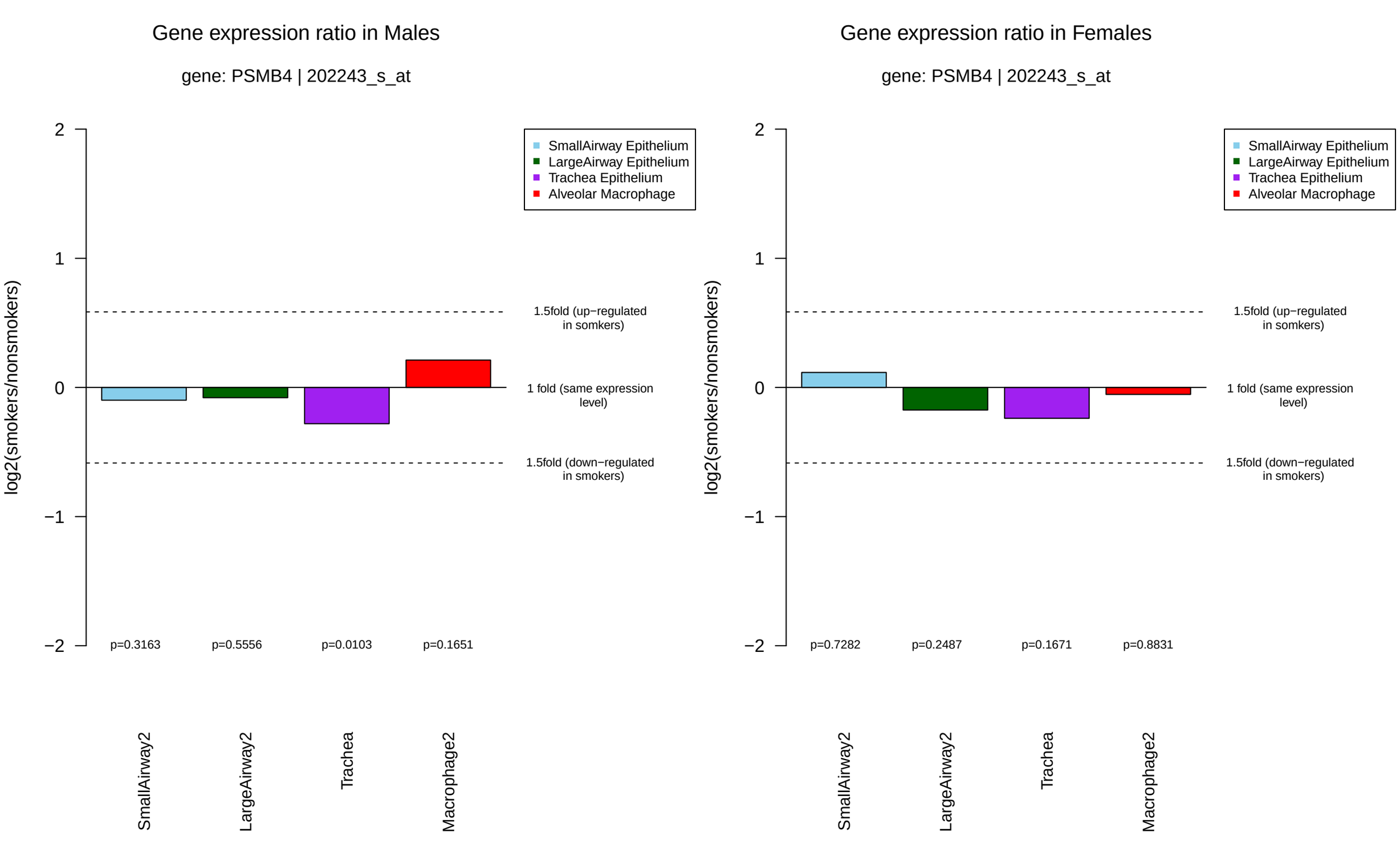
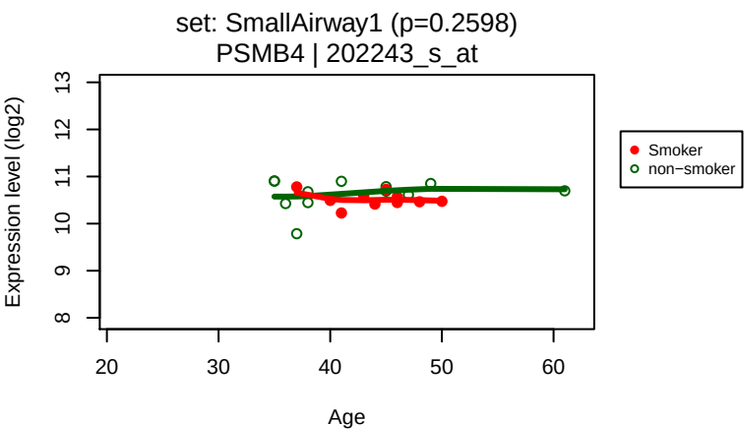
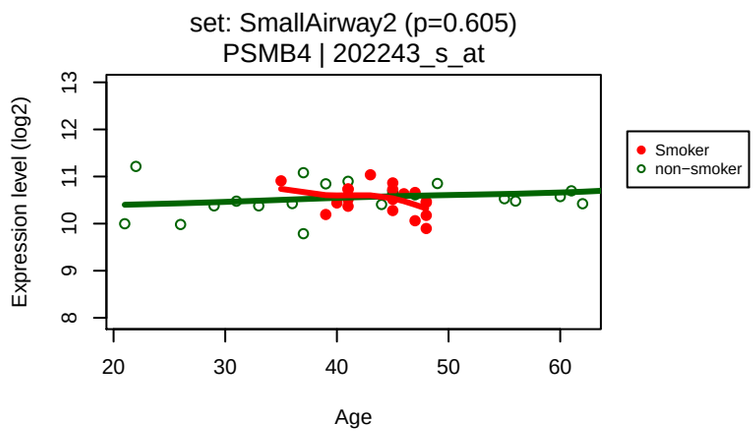
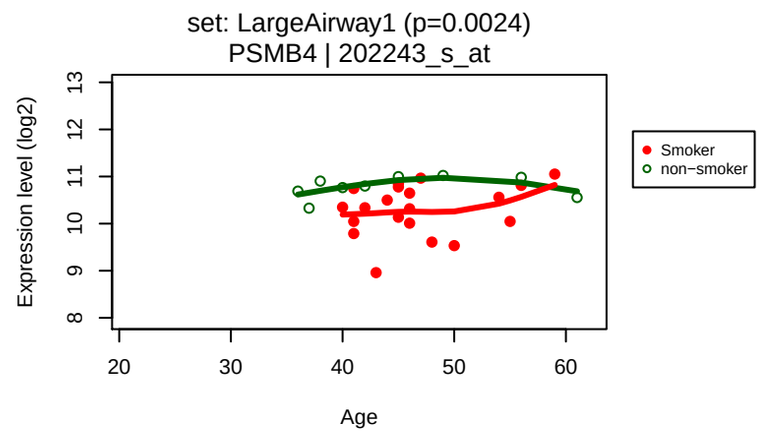

Gene expression of smokers and nonsmokers
gene: PSMB4 | 202243_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2598
p=0.605
p=0.1136
p=0.0024
p=0.2586
p=0.0094
p=0.0209
p=0.4174
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

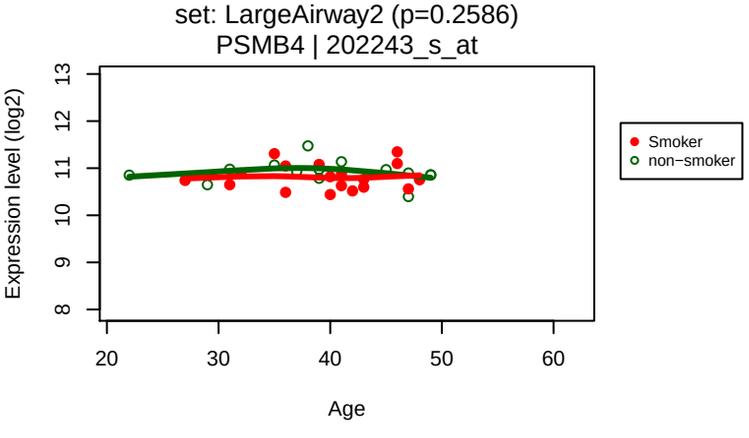

Gene expression ratio (smokers/nonsmokers)
gene: PSMB4 | 202243_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2598
p=0.605
p=0.1136
p=0.0024
p=0.2586
p=0.0094
p=0.0209
p=0.4174
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

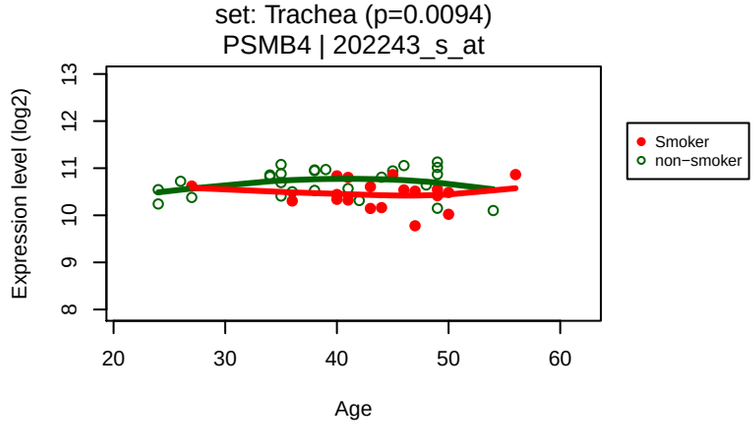

Gene expression ratio in Males
gene: PSMB4 | 202243_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3163
p=0.5556
p=0.0103
p=0.1651
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMB4 | 202243_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7282
p=0.2487
p=0.1671
p=0.8831
SmallAirway2
LargeAirway2
Trachea
Macrophage2

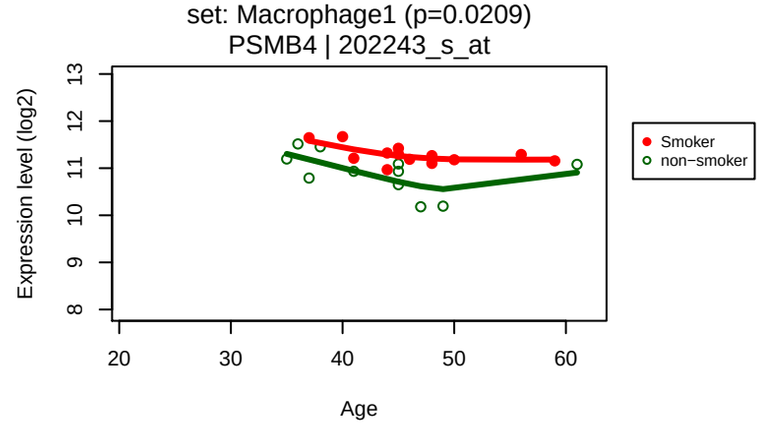

set: SmallAirway1 (p=0.2598)
PSMB4 | 202243_s_at
Expression level (log2)
Age
Smoker
non−smoker

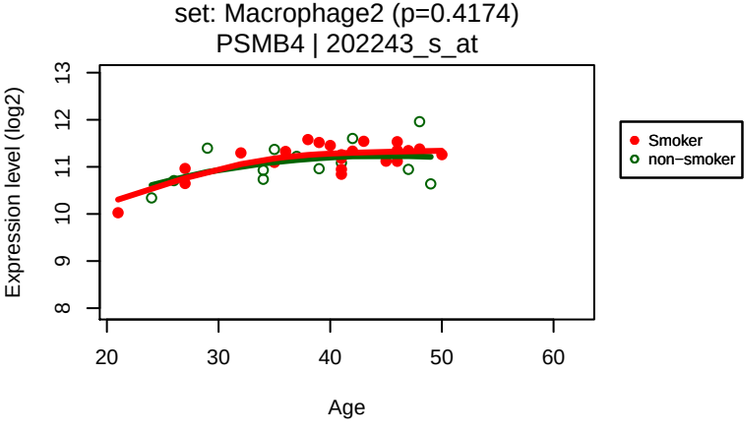

set: SmallAirway2 (p=0.605)
PSMB4 | 202243_s_at
Expression level (log2)
Age
Smoker
non−smoker

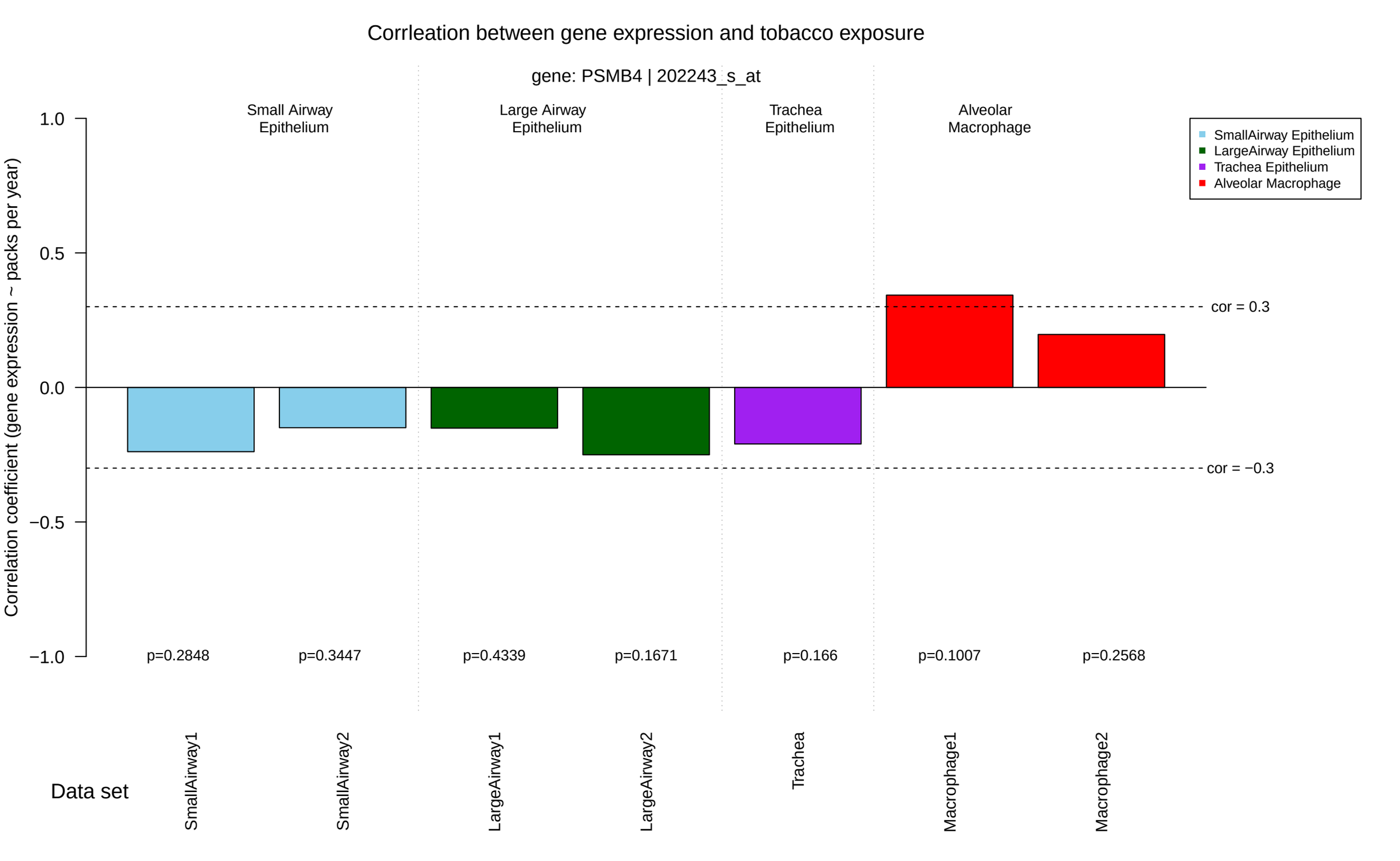

set: LargeAirway1 (p=0.0024)
PSMB4 | 202243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2586)
PSMB4 | 202243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0094)
PSMB4 | 202243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0209)
PSMB4 | 202243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4174)
PSMB4 | 202243_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMB4 | 202243_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2848
p=0.3447
p=0.4339
p=0.1671
p=0.166
p=0.1007
p=0.2568
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2